

1 Supplement — Construction of the *Escherichia coli* and
 2 *Methanococcus maripaludis* etc-GEMs, the Sharpe–Schoolfield
 3 E_a audit, and robustness

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20 This Supplement documents the full construction of **both** etc-GEMs — the *E. coli* respiration
 21 model and the *M. maripaludis* methanogenesis model, each the first published for its taxon
 22 — together with the window-independence audit that motivates the Sharpe–Schoolfield E ,
 23 the calibration diagnostics, and the robustness analyses. Numbers are model outputs; honest
 24 caveats are stated throughout.

1 Building the *E. coli* respiration etc-GEM

Base network and enzyme constraint. The base is iML1515, the manually curated *E. coli* K-12 MG1655 genome-scale reconstruction (2,712 reactions, 1,877 metabolites, 1,516 genes)¹, converted to the enzyme-constrained ecModel **eciML1515** with the GECKO method^{2,3}, which adds a protein-usage pseudo-metabolite per reaction drawn from a shared pool. The metabolic pool is set from independent data ($P_{\text{metab}} = \sigma f_{\text{metab}} P^{\text{lit}}$; literature protein content $P^{\text{lit}} = 0.5$ g gDW⁻¹, measured metabolic fraction, in-vivo saturation $\sigma = 0.45^{4,5}$), so the growth-rate magnitude is emergent, not fit.

Model curation — uncosted O sinks. Four reactions in eciML1515 consume O at negligible enzyme cost — two quinol/menaquinol monooxygenases (QMO2, QMO3), a malate:O oxidoreductase (MOX) and a cuprous oxidase (CU1Opp) — forming a futile O cycle that at low temperature can carry the majority of gross O turnover and short-circuits the enzyme-costed cytochrome oxidase. Closing these four (a documented, reversible default) routes O through the genuine respiratory chain at $\sim -0.3\%$ growth, so control is attributed to real respiration. Catalase and superoxide dismutase are real enzymes and are deliberately *not* closed.

Thermal + allocation layers. Per-enzyme T_{opt} and T_m were grounded from a sequence-based optimum predictor⁶ and meltome/length relations for $\sim 90\%$ of enzymes (the rest at dataset means), with DLTkcat⁷ refining T_{opt} where it fits; $k_{\text{cat}}(T)$ follows MMRT⁸ with a literature ΔC_p prior, and the high- T collapse follows a two-state unfolding native fraction keyed on T_m . The proteome was partitioned into metabolic/biosynthesis/maintenance sectors^{9,10} from a growth-condition-matched measured *E. coli* proteome, with the coupled bacterial growth law (Scott slope) under which the ribosomal fraction rises with growth rate. Maintenance is a temperature-dependent NGAM(T).

Validation and calibration. The emergent model was validated against, then Bayesian-calibrated (emcee, exact Gaussian discrepancy likelihood) to, the rich-medium (BHI) growth TPC of¹¹ (7–46 °C, $r_{\text{max}} = 2.40$ h⁻¹ at 40 °C). The calibration frees only a few global scalars (the grounded per-enzyme parameters are not re-fit) and closes $\sim 89\%$ of the magnitude gap; the posterior in-vivo saturation railed toward its physical ceiling ($\sigma = 0.867$ [0.671, 0.986], above the 0.4–0.5 literature range — a genuine residual, kept distinct rather than papered over), with a melting-temperature shift ($dT_m = -5.6$ K) pulling the hot collapse to the data (Figure 1). The calibrated E_a holds near the metabolic benchmark. This is the calibrated model dissected in the main text.

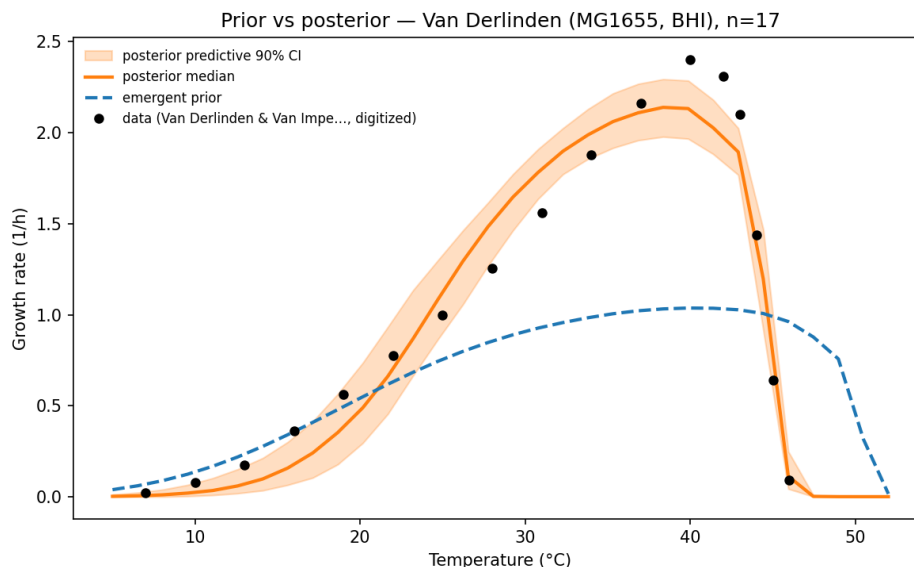


Fig. 1 | Prior (emergent) vs posterior *E. coli* TPC against the Van Derlinden 2012 BHI data.

2 Building the *M. maripaludis* methanogenesis etc-GEM (M1–M6)

The model was built in six phases, each committed and validated (build records in `strains/mmaripaludis/outputs/M*.md`).

2.1 M1 — base GEM + medium (foundation)

The iMR539 reconstruction¹² (688 reactions / 710 metabolites / 539 genes; BioModels BIOMD0000001099) was acquired and validated on a defined H₂/CO₂ medium. Plain FBA reproduces hydrogenotrophic methanogenesis with the correct **H₂:CO₂:CH₄ = 4:1:1** stoichiometry and a gene-annotated Wolfe cycle. A preliminary audit found **no thermodynamic energy loops**; 85% metabolic GPR coverage; MMP locus tags map to UniProt proteome UP000000590 (sequences fetchable) — ecModel-ready.

2.2 M1b — carbon-honest curation

The published reconstruction could not grow without free uptake of four biomass precursors. Tracing the network, three (Membrane_lipid, NAC, Flagellin) feed *only* the archaellin (flagellum) glycoprotein — a dispensable motility structure — so archaellin was folded out of biomass; the fourth (tRNA(SeCys)) was a **recycling gap** (the incorporation step failed to release the free tRNA), fixed by regenerating it. Organic-carbon leaks were closed and the pinned

73 CH₄ exchange freed. The curated model grows autotrophically on H₂/CO₂ with **99.98% of**
 74 **biomass carbon from CO₂**. Two vitamin cofactors (thiamin, NMN) lack biosynthesis and
 75 are supplied as documented medium traces (0.02% of carbon).

76 2.3 M2 / M2b — the sMOMENT ecModel + kcat refinement

77 An sMOMENT total-protein pool was attached (reusing the E. coli `enzyme_cost` engine):
 78 each reaction costs MW/k_{cat} from a shared budget. Per-reaction k_{cat} by source priority:
 79 **measured core** (electron-bifurcation complexes from Milton et al. 2018, 25–165 U/mg¹³;
 80 classic biochemistry for Mtr/Fwd/Mcr/etc.), **DLTKcat** sequence predictions for the bulk, and
 81 a mean fallback. Mcr — the famously slow terminal enzyme and the E_a target — was sourced
 82 carefully (mature-Mcr average specific activity ~20 U/mg, *Methanothermobacter*; k_{cat} 59 s⁻¹,
 83 uncertainty range 3–294 s⁻¹). **M2b** corrected a systematic DLTKcat archaeal-underprediction
 84 (28% of predictions <1 s⁻¹) with cited literature/BRENDA overrides for the flagged high-pool
 85 offenders (pyruvate- and 2-oxoglutarate:ferredoxin oxidoreductases) plus a documented 1 s⁻¹
 86 floor. The grounded pool ($P_{total} \times \sigma \times f_{metab}$, not tuned to μ_{max}) gives an emergent μ that
 87 undershoots the observed rate ~4×, expected for a-priori kinetics.

88 2.4 M3 — thermal envelope + measured maintenance

89 $kcat(T)$ (MMRT/Eyring) $\times$ native fraction $f_N(T)$ (two-state unfolding keyed on T_m), with per-
 90 enzyme optima from a documented **mesophile Topt prior** and T_m from a mesophile prior (the
 91 sequence/meltome predictors are the future upgrade). Maintenance is a temperature-dependent
 92 NGAM(T) **anchored on the measured** M. maripaludis maintenance energy (7.836 mmol
 93 ATP gDW⁻¹ h⁻¹, Goyal et al. 2015¹⁴). The emergent TPC matches Jones' **CTmax (~47 °C)**
 94 a-priori (Figure 2); the rising limb was too warm/steep and the magnitude maintenance-crushed
 95 — the residuals the calibration then addresses.

96 2.5 M4 — Bayesian calibration to Jones 1983

97 The thermal ecModel was calibrated to the digitised Jones 1983 growth TPC¹⁵ by ensemble
 98 MCMC (emcee; the same machinery as the E. coli fit), magnitude-first, with provenance priors.
 99 The posterior **reproduces the whole curve** (Figure 3; RMSE 0.005 h⁻¹), reaching the peak
 100 (0.184 vs 0.181 h⁻¹) with Topt 37 °C and CTmax 51 °C. The demanded magnitude correction
 101 ($kcat_scale$ 7×) is a plausible in-vitro→in-vivo + archaeal-underprediction gap (the single
 102 sMOMENT pool lumps the degenerate magnitude levers). The maintenance multiplier stayed
 103 near 1, confirming the Goyal-measured NGAM.

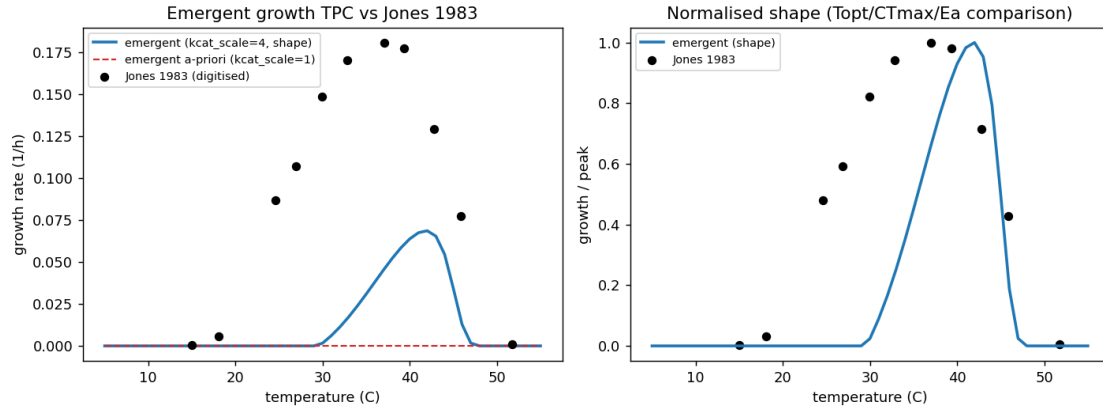


Fig. 2 | Emergent (nothing-fit) methanogen TPC vs Jones 1983: the falling limb (CTmax) matches a-priori; the rising limb and magnitude are calibrated in M4.

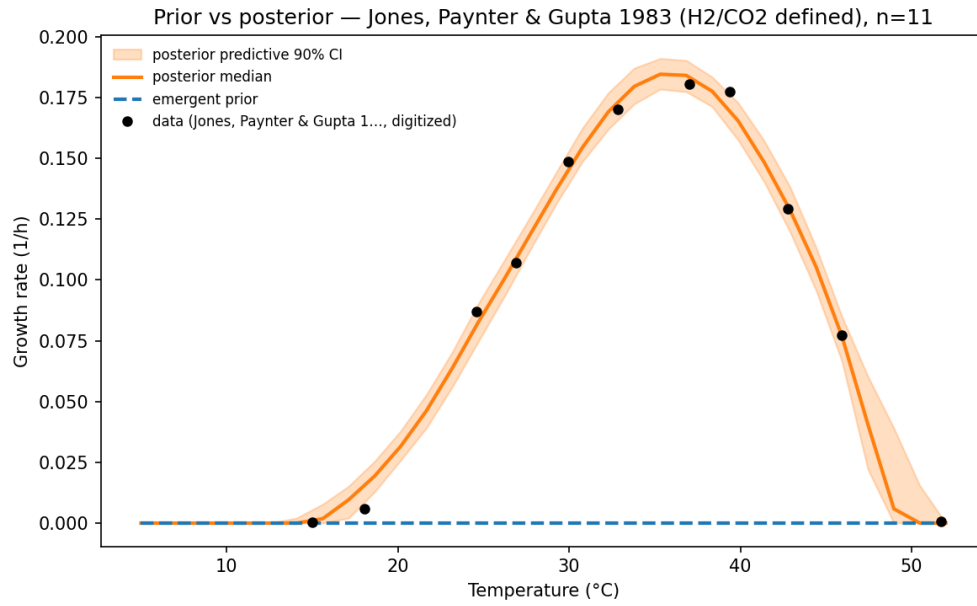


Fig. 3 | Prior (emergent) vs posterior methanogen TPC against Jones 1983.

2.6 M6 — the methanogen’s own allocation layer

A proteome-sector/growth-law layer was added, grounded in the organism’s **own** allocation physiology: Müller et al. 2021¹⁶ show *M. maripaludis* holds a **constant ribosomal proteome fraction** and invariant catabolic/anabolic allocation across growth rate (the opposite of the *E. coli*/yeast scaling law), so the growth-law coupling slope is **~0 (flat)**. Sector fractions (f_{metab} 0.55 / f_{maint} 0.30 / f_{bio} 0.15) are grounded in the Xia/Müller proteome. Forward-check: with the sector layer on and the M4 posterior, the TPC is **identical** to the single-pool model ($\max|\Delta\mu| = 0$), so no re-calibration was needed. The directly-measured allocation buffer is **+0.000 eV** (vs *E. coli* −0.130), grounding the allocation asymmetry in each organism’s measured strategy.

Honest caveats carried into the analysis. (i) The **Mcr** k_{cat} (3–294 s^{−1}) is the dominant single-parameter uncertainty. (ii) **DLTKcat is bacteria-trained**, so archaeal predictions are corrected but remain a prior for the biosynthetic bulk. (iii) The T_m **and Topt** are mesophile priors (no archaeal sequence predictor). (iv) The **sector fractions** are grounded but approximate; the flat slope is the robust, cited quantity. The main-text conclusions are shown robust to the parameter that matters most (Mcr; Section 5).

3 Parameter provenance and the global correction knobs

Every per-enzyme and whole-cell input is grounded in independent data or a stated literature value; nothing is fit to the growth curve in the emergent model. Table 1 and Table 2 give the provenance for each organism, and Table 3 the global scalars freed in calibration (each transforms the whole grounded per-enzyme distribution while preserving its shape; all are no-ops at their nominal).

Table 1: *E. coli* parameter provenance — grounded in independent data, never fit to growth.

Symbol	Quantity	Source	Coverage / value
$MW_i, k_{\text{cat,ref},i}$	enzyme mass, reference turnover	GECKO ecModel of iML1515 ^{1–3}	2,560 enzymes
$T_{\text{opt},i}$	catalytic optimum	sequence predictor ⁶ + DLTKcat ⁷	~90% grounded
$T_{m,i}$	melting temperature	measured <i>E. coli</i> meltome ¹⁷	~90% grounded; rest 55.6 °C
$\Delta C_{p,i}^\ddagger$	turnover curvature	literature MMRT prior ⁸	−4 kJ mol ^{−1} K ^{−1}
$f_{\text{metab}}/f_{\text{bio}}/f_{\text{maint}}$	sector fractions	measured proteome per medium & T ¹⁸	0.48/0.19/0.33 (glucose 30 °C)

Symbol	Quantity	Source	Coverage / value
s	growth-law slope	Scott 2010 ¹⁰	0.33 (rising ribosome fraction)
P^{lit}, σ	total protein, saturation	literature ^{4,5}	0.5 g gDW ⁻¹ , 0.45
NGAM(T)	maintenance	<i>E. coli</i> maintenance port ¹⁹	a_m 8.5, E_m 0.5 eV
validation	growth TPC	K-12 MG1655, BHI, 7–46 °C ¹¹	1 curve

Table 2: *M. maripaludis* parameter provenance. The measured energy-core kinetics, the measured maintenance, and the constant-ribosome allocation law are the organism-specific grounded inputs.

Symbol	Quantity	Source	Coverage / value
MW_i	enzyme mass	UniProt proteome UP000000590 (from GPRs)	527/527 enzymatic reactions
$k_{\text{cat},i}$	turnover	measured core ¹³ + DLTkcat ⁷ + literature overrides + 1 s ⁻¹ floor	11 core / 293 DLTkcat / 223 fallback
$k_{\text{cat}}(\text{Mcr})$	terminal enzyme	mature-Mcr specific activity	58.9 s ⁻¹ (range 3–294; swept)
$T_{\text{opt},i}, T_{m,i}$	optimum, stability	mesophile prior (sequence/meltome predictors not run)	T_{opt} ~39 °C, T_m ~55.6 °C
$\Delta C_{p,i}^\ddagger$	turnover curvature	literature MMRT prior ⁸	−4 kJ mol ⁻¹ K ⁻¹
$f_{\text{metab}}/f_{\text{bio}}/f_{\text{maint}}$	sector fractions	Xia/Müller proteome ¹⁶	0.55/0.15/0.30
s	growth-law slope	Müller 2021 constant ribosome fraction ¹⁶	0 (flat law)
P^{lit}, σ	total protein, saturation	literature ^{4,5}	0.5 g gDW ⁻¹ , 0.45
NGAM(T)	maintenance	measured <i>M. maripaludis</i> ¹⁴	7.836 mmol ATP gDW ⁻¹ h ⁻¹ @ 37 °C
validation	growth TPC	type strain JJ, H /CO ¹⁵	1 curve

Table 3: Global correction knobs freed in calibration — each is one scalar acting on the whole per-enzyme distribution.

Knob	Transformation	Effect
dT_{opt}	$T_{\text{opt},i} \rightarrow T_{\text{opt},i} + dT_{\text{opt}}$	uniform shift of every optimum
<code>topt_scale</code>	$T_{\text{opt},i} \rightarrow T_0 + \text{topt_scale}(T_{\text{opt},i} - T_0)$	spread of optima
<code>dCp_scale</code>	$\Delta C_{p,i}^{\ddagger} \rightarrow \text{dCp_scale} \Delta C_{p,i}^{\ddagger}$	turnover curvature / breadth
dT_m	evaluate $f_{N,i}$ at $T - dT_m$	uniform melting shift (+ raises CT_{max})
<code>tm_scale</code>	$T_{m,i} \rightarrow \bar{T}_m + \text{tm_scale}(T_{m,i} - \bar{T}_m)$	spread of melting temperatures
<code>kcat_scale</code> (λ_k)	$c_i \rightarrow c_i / \lambda_k$	global turnover level $\rightarrow r_{\text{max}}$
$f_{\text{metab}}, f_{\text{maint}}$	set the sector simplex	proteome allocation
<code>ngam_scale</code> , <code>ngam_steepness</code>	scale a_m, E_m	maintenance amplitude / steepness
σ	scale both sector caps	in-vivo saturation (magnitude)
σ_{disc}	likelihood variance	model-discrepancy noise (nuisance)

4 Why Sharpe–Schoolfield E : the window-sensitivity audit

The naive Boltzmann–Arrhenius slope of the rising limb depends on the fitting window because the limb is curved. Table 4 recomputes it under five window conventions for every model and observed curve: the spread is **0.42–1.41 eV per curve**, larger than the effects we attribute, and enough to reverse the cross-organism ordering. The Sharpe–Schoolfield E (the high-temperature-inactivation model, main text) is read from the whole shape, is window-independent, and — the decisive test — makes model and observed agree for both organisms (Figure 4).

```
import pandas as pd
w = pd.read_csv("assets/tables/window_sensitivity.csv")
ss = pd.read_csv("assets/tables/sharpe_schoolfield_fits.csv")[["curve", "SS_E_eV", "SS_r2" if
m = w.merge(ss, on="curve", how="left")
print(m.round(3).to_markdown(index=False))
```

Table 4: Full window-sensitivity table: Boltzmann-Arrhenius slope E_a (eV) under five window conventions, and the window-independent Sharpe-Schoolfield E , for every model + observed curve.

curve	full_be-					spread_eV	SS_E_eV	r2
	rel_10_95	low	rel_10_80	rel_30_90	rel_10_50			
ecoli_emer- gent	0.609	0.64	0.802	0.489	1.054	0.565	0.458	0.896
ecoli_tuned	0.956	1.468	1.203	0.756	1.56	0.804	0.677	0.93
ecoli_ob- served_VdL	0.643	0.963	0.683	0.545	0.839	0.418	0.558	0.973
methan_cal- ibrated	1.174	1.462	1.372	0.892	1.785	0.893	1.024	0.957
methan_ob- served_Jones	0.66	2.067	nan	0.781	nan	1.407	1.042	0.95

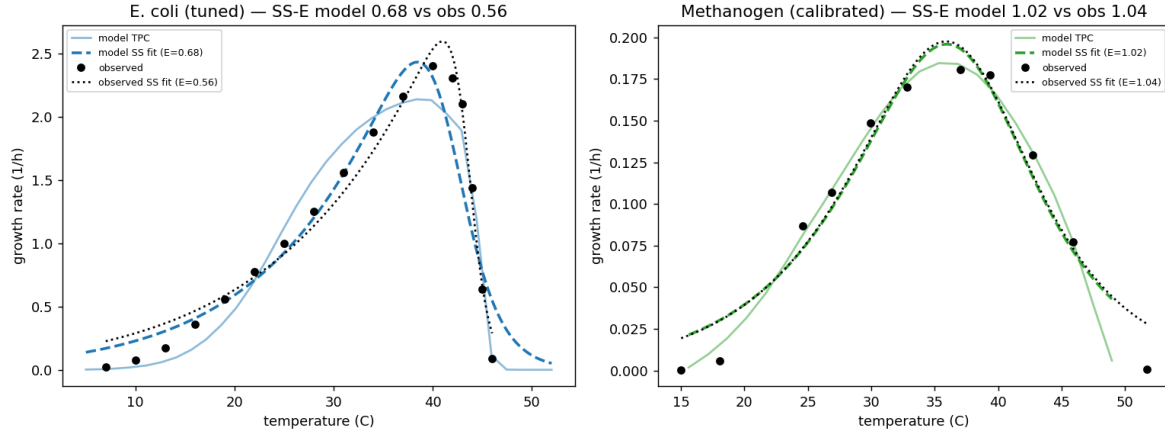


Fig. 4 | Sharpe–Schoolfield fits overlaid on the model and observed curves for both organisms; the model and observed E agree.

5 Mcr- k_{cat} sensitivity

Table 5 sweeps Mcr across its 3–294 s⁻¹ range. The organism E_a stays **1.01–1.27 eV**, always well above **E. coli’s 0.68**, and the methanogenesis backbone carries **83–103%** of the control term throughout — so the ordering and the backbone-control finding are robust. Only the identity of the single leading enzyme shifts (Mcr when slow → Fwd when fast).

```
import pandas as pd
m = pd.read_csv("assets/tables/mcr_sweep.csv")
print(m.round(3).to_markdown(index=False))
```

Table 5: Mcr-kcat sensitivity sweep: organism SS-E, backbone fraction of the control term, and the leading enzymes, across 3-294/s.

Mcr_kcat	SS_E	slope	back- bone_frac_of_con- trol	top3	Mcr_C_i	Mcr_con- trib
3	1.274	1.597	1.026	Mcr(1.605);Fwd(0.092);CODH(0.065)	1.605	1.605
10	1.01	1.105	0.986	Mcr(0.876);Fwd(0.167);CODH(0.039)	0.876	0.876
30	1.065	1.132	0.925	Mcr(0.488);Fwd(0.277);Mtr(0.059)	0.488	0.488
58.9	1.056	1.081	0.887	Fwd(0.337);Mcr(0.304);Mtr(0.077)	0.304	0.304
100	1.066	1.153	0.857	Fwd(0.395);Mcr(0.211);Mtr(0.078)	0.211	0.211
294	1.017	1.074	0.833	Fwd(0.427);Mtr(0.086);Mer(0.076)	0.076	0.078

6 Calibration diagnostics

Both models were calibrated by ensemble Markov-chain Monte Carlo (emcee) with an exact Gaussian discrepancy likelihood, magnitude-first initialisation and autocorrelation-based early stopping; the grounded per-enzyme parameters are held fixed and only a small set of global scalars is freed. Figure 7 shows the posterior corner plots. For *E. coli* the fit demands a large in-vivo saturation ($\sigma \rightarrow$ its physical ceiling) and a modest melting-temperature shift; for *M. maripaludis* it demands a several-fold turnover magnitude correction (kcat_scale = 7, absorbing the in-vitro→in-vivo gap and the residual archaeal-kcat underprediction on the single sMOMENT pool), while the maintenance multiplier stays near one — confirming the Goyal-measured NGAM. In both cases the posterior-predictive curve passes through the observed data (Fig. 4 and the main-text Figure 1), and the resulting Sharpe–Schoolfield E_a matches the observed value.

- Monk, J. M. *et al.* [iML1515, a knowledgebase that computes Escherichia coli traits](#). *Nature Biotechnology* **35**, 904–908 (2017).
- Sánchez, B. J. *et al.* [Improving the phenotype predictions of a yeast genome-scale metabolic model by incorporating enzymatic constraints \(GECKO\)](#). *Molecular Systems Biology* **13**, 935 (2017).
- Domenzain, I. *et al.* [Reconstruction of a catalogue of genome-scale metabolic models with enzymatic constraints using GECKO 2.0](#). *Nature Communications* **13**, 3766 (2022).

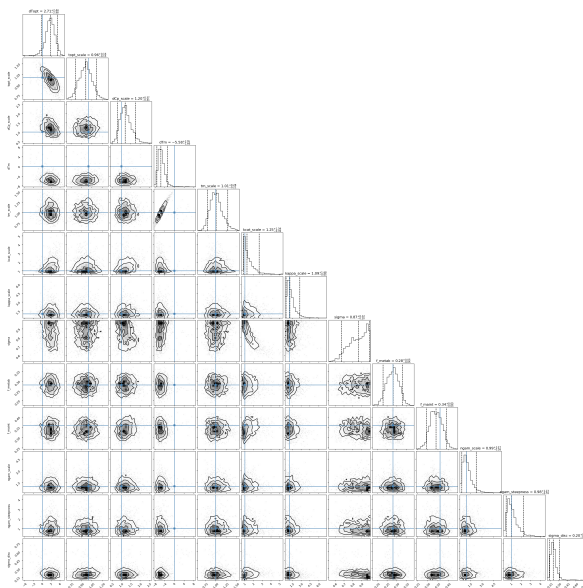


Fig. 5 | *E. coli* (Van Derlinden 2012).

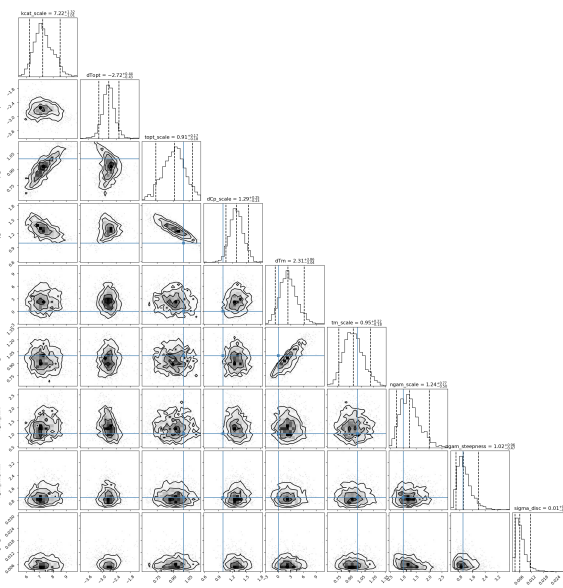


Fig. 6 | *M. maripaludis* (Jones 1983).

Posterior corner plots for the two calibrations.

Fig. 7

4. Davidi, D. *et al.* [Global characterization of in vivo enzyme catalytic rates and their correspondence to in vitro kcat measurements](#). *Proceedings of the National Academy of Sciences* **113**, 3401–3406 (2016).
5. Heckmann, D. *et al.* [Kinetic profiling of metabolic specialists demonstrates stability and consistency of in vivo enzyme turnover numbers](#). *Proceedings of the National Academy of Sciences* **117**, 23182–23190 (2020).
6. Li, G., Rabe, K. S., Nielsen, J. & Engqvist, M. K. M. [Machine learning applied to predicting microorganism growth temperatures and enzyme catalytic optima](#). *ACS Synthetic Biology* **8**, 1411–1420 (2019).
7. Qiu, S., Zhao, S. & Yang, A. [DLTKcat: Deep learning-based prediction of temperature-dependent enzyme turnover rates](#). *Briefings in Bioinformatics* **25**, bbad506 (2024).
8. Hobbs, J. K. *et al.* [Change in heat capacity for enzyme catalysis determines temperature dependence of enzyme catalyzed rates](#). *ACS Chemical Biology* **8**, 2388–2393 (2013).
9. Basan, M. *et al.* [Overflow metabolism in Escherichia coli results from efficient proteome allocation](#). *Nature* **528**, 99–104 (2015).
10. Scott, M., Gunderson, C. W., Mateescu, E. M., Zhang, Z. & Hwa, T. [Interdependence of cell growth and gene expression: Origins and consequences](#). *Science* **330**, 1099–1102 (2010).

11. Van Derlinden, E. & Van Impe, J. F. Quantifying the heterogeneous heat response of *Escherichia coli* under dynamic temperatures. *International Journal of Food Microbiology* **158**, 73–78 (2012).
12. Richards, M. A. *et al.* Exploring hydrogenotrophic methanogenesis: A genome scale metabolic reconstruction of *Methanococcus maripaludis*. *Journal of Bacteriology* **198**, 3379–3390 (2016).
13. Milton, R. D. *et al.* Measured specific activities and redox thermodynamics of the flavin-based electron-bifurcation hydrogenase and heterodisulfide-reductase complexes of *Methanococcus maripaludis*. *Biochemistry* (2018).
14. Goyal, N., Widiastuti, H., Karimi, I. A. & Zhou, Z. Flux measurements and maintenance energy for carbon dioxide utilization by *Methanococcus maripaludis*. *Microbial Cell Factories* **14**, 146 (2015).
15. Jones, W. J., Paynter, M. J. B. & Gupta, R. Characterization of *Methanococcus maripaludis* sp. Nov., a new methanogen isolated from salt marsh sediment. *Archives of Microbiology* **135**, 91–97 (1983).
16. Müller, A. L. *et al.* An alternative resource allocation strategy in the chemolithoautotrophic archaeon *Methanococcus maripaludis*. *Proceedings of the National Academy of Sciences* **118**, e2025854118 (2021).
17. Leuenberger, P. *et al.* [Cell-wide analysis of protein thermal unfolding reveals determinants of thermostability](#). *Science* **355**, eaai7825 (2017).
18. Wang, D. *et al.* A proteome optimal allocation model for elucidating effects of temperature on bacterial growth. *npj Systems Biology and Applications* <https://doi.org/10.1038/s41540-026-00693-4> (2026) doi:10.1038/s41540-026-00693-4.
19. Madkaikar, A. Predicting the temperature dependence of microbial metabolism with enzyme- and temperature-constrained genome-scale models. (Imperial College London, 2023).